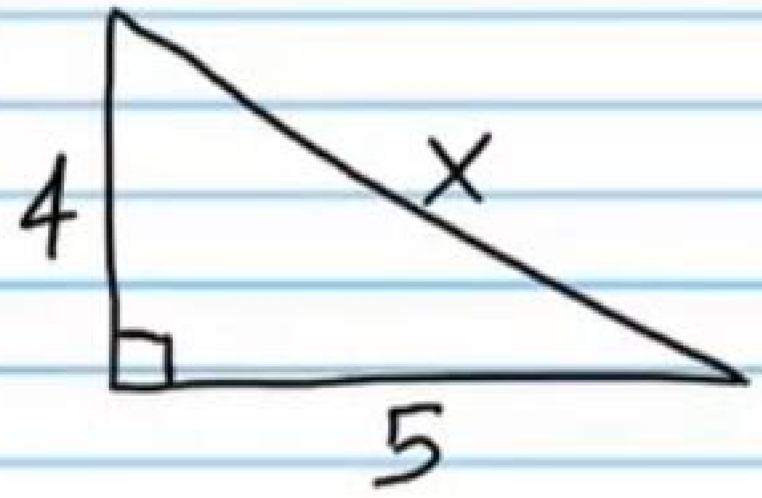
Name
Date
Course

Homework

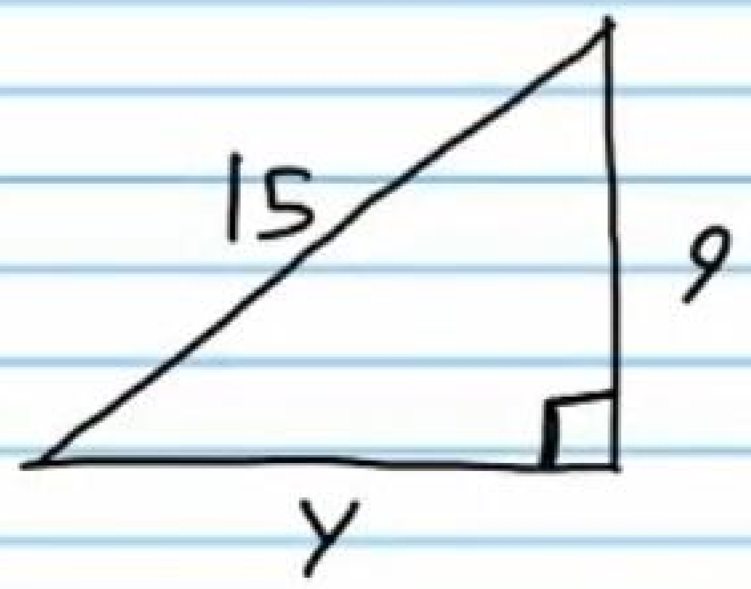
Find the missing side of each right triangle.

①



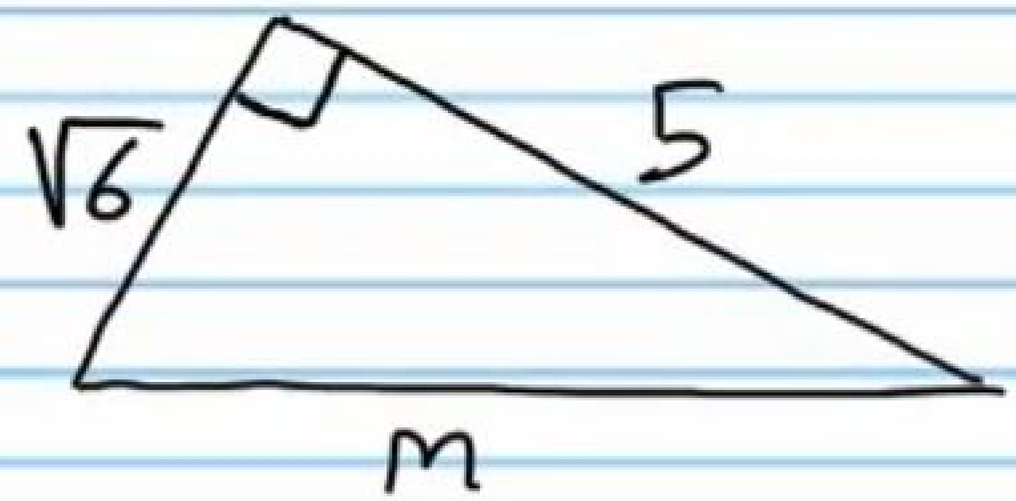
$X = \sqrt{41}$

②



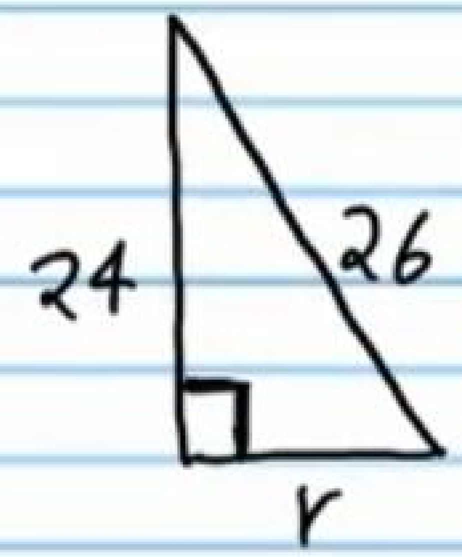
$Y = 12$

③



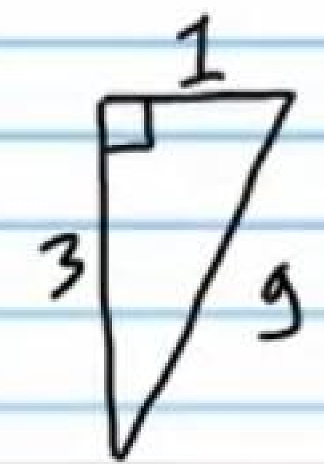
$m = \sqrt{31}$

④



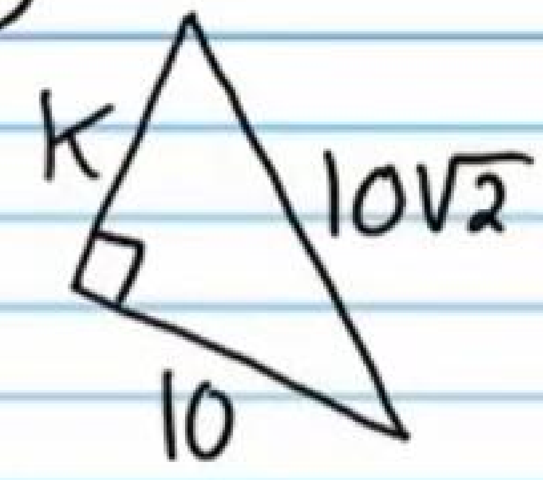
$r = 10$

⑤



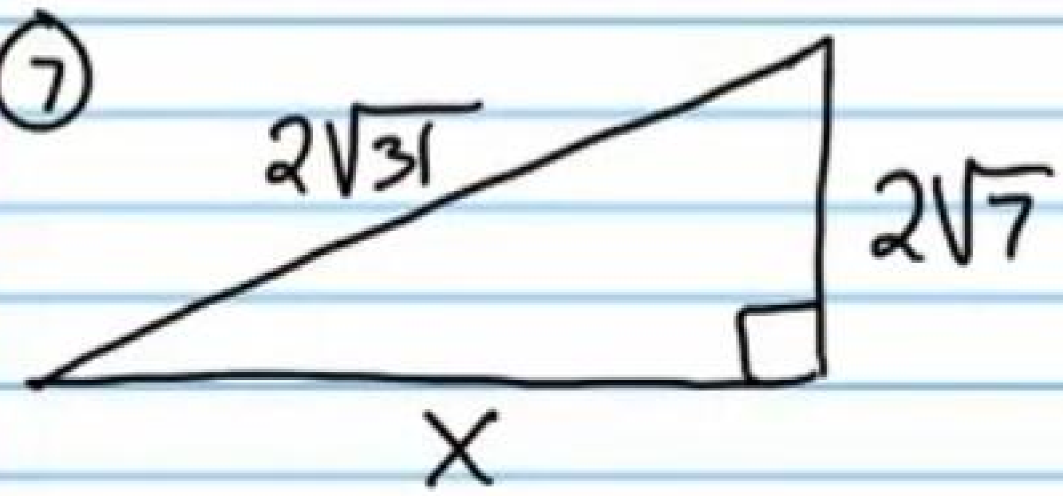
$g = \sqrt{10}$

⑥



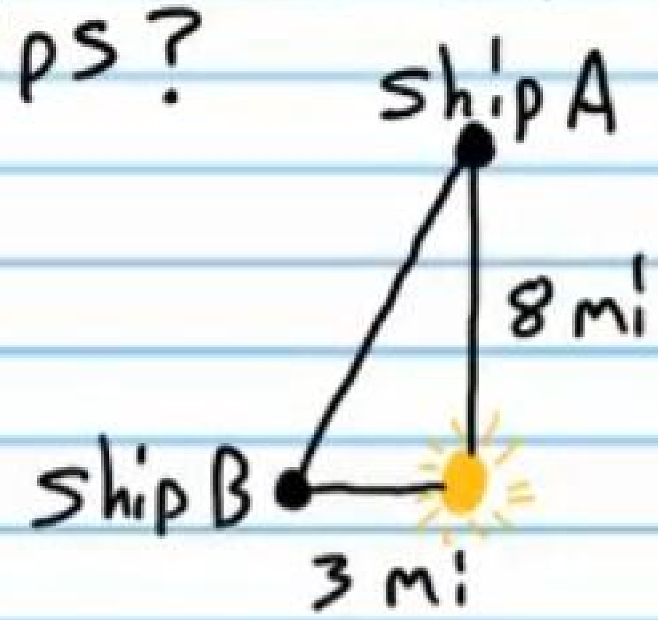
$K = 10$

⑦



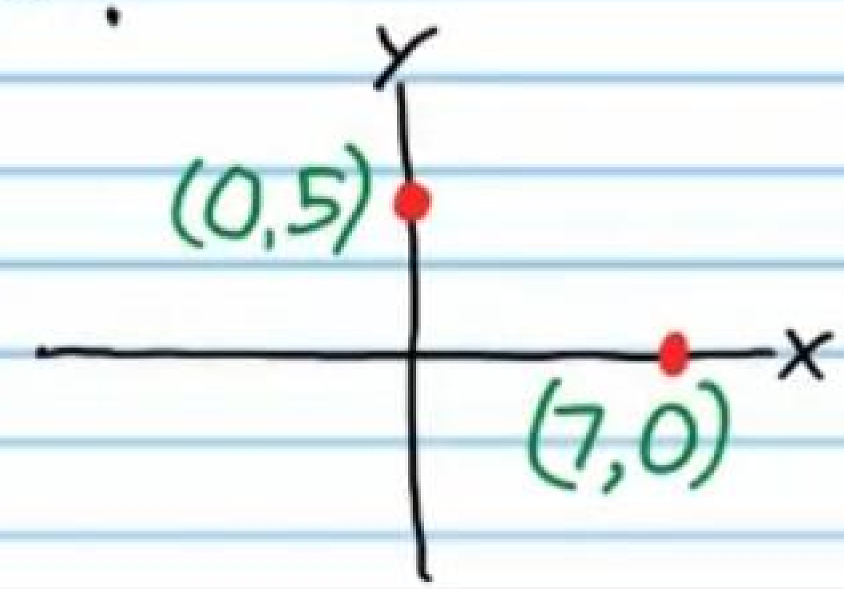
$X = 4\sqrt{6}$

⑧ A ship is 8 miles north of a lighthouse. Another ship is 3 miles west of the same lighthouse. How far apart are the ships?



$\sqrt{73}$ mi

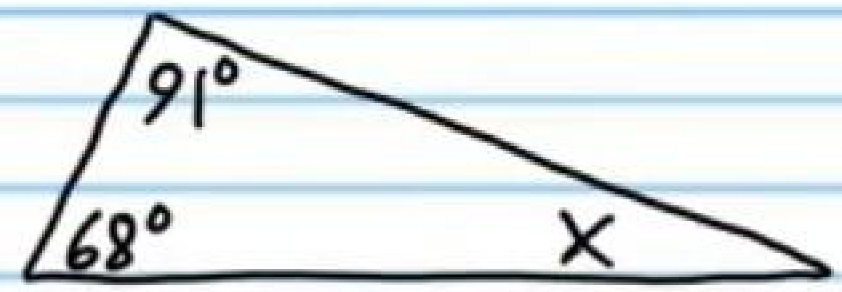
⑨ What is the distance between the two coordinates below?



$\sqrt{74}$

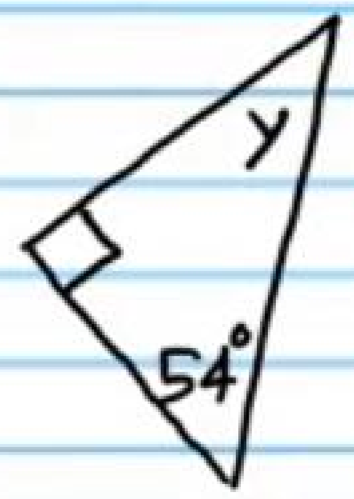
Find the value of each variable.

(10)



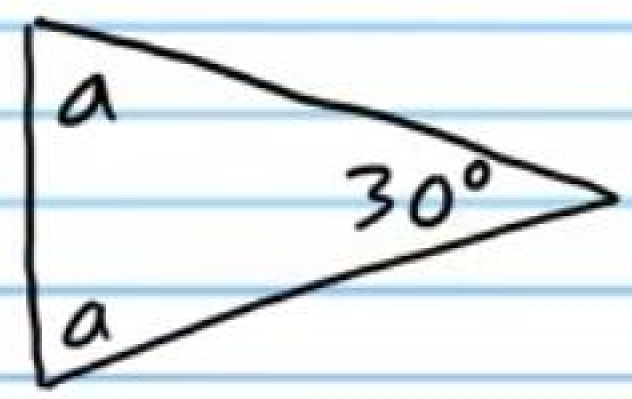
$\boxed{x = 21°}$

(11)



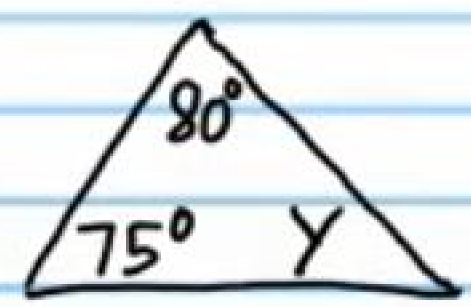
$\boxed{y = 36°}$

(12)



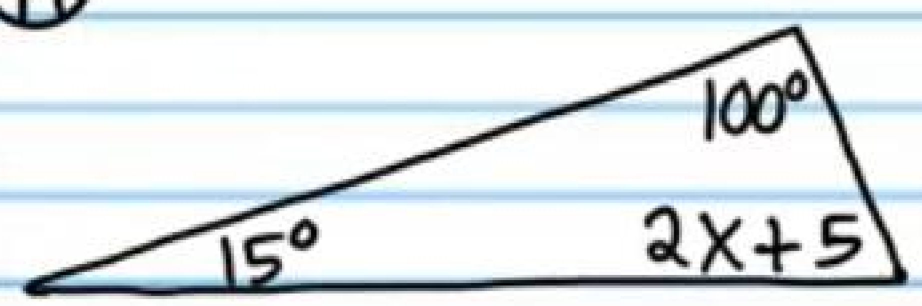
$\boxed{a = 75°}$

(13)



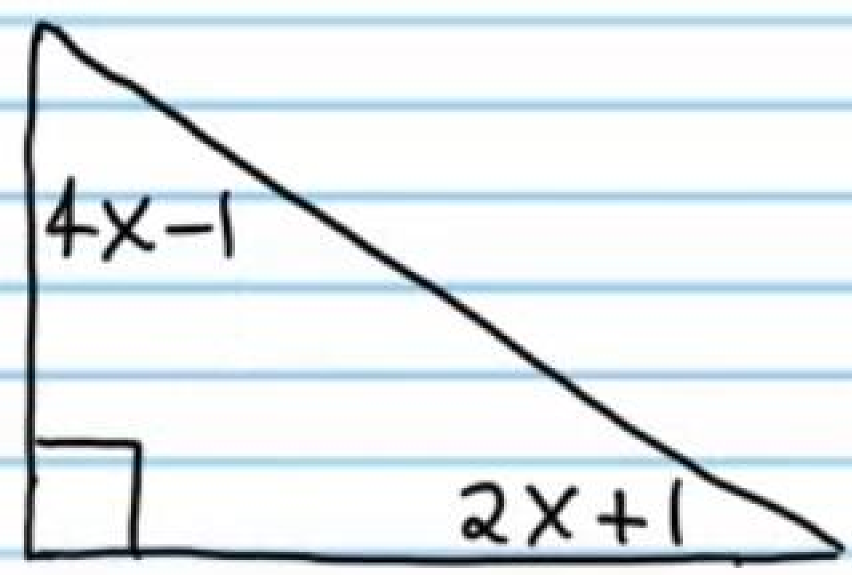
$\boxed{y = 25°}$

(14)



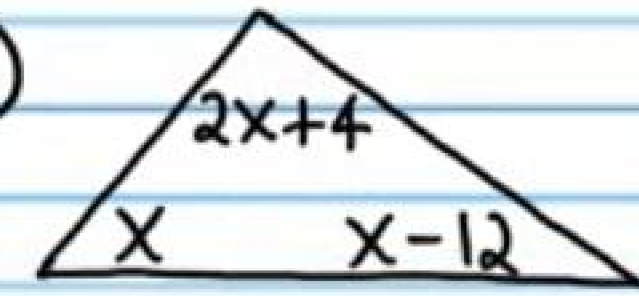
$\boxed{x = 30}$

(15)



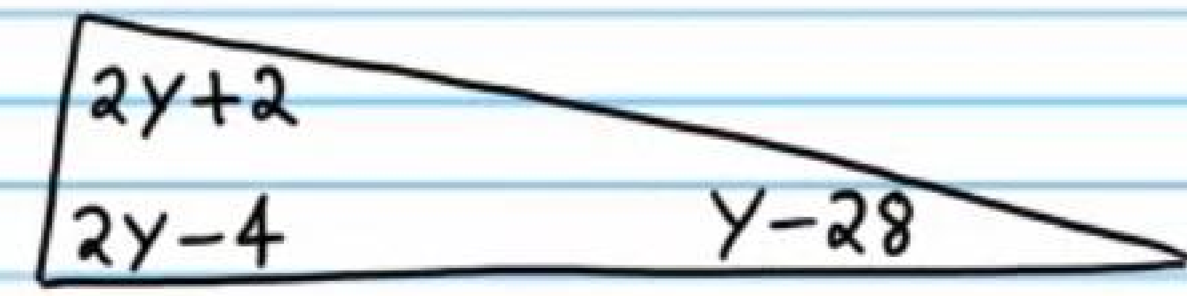
$\boxed{x = 15}$

(16)



$\boxed{x = 47°}$

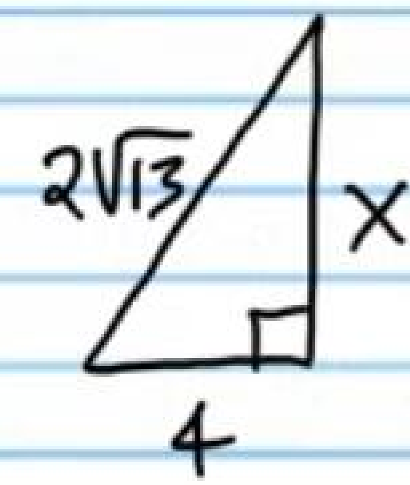
(17)



$\boxed{y = 42}$

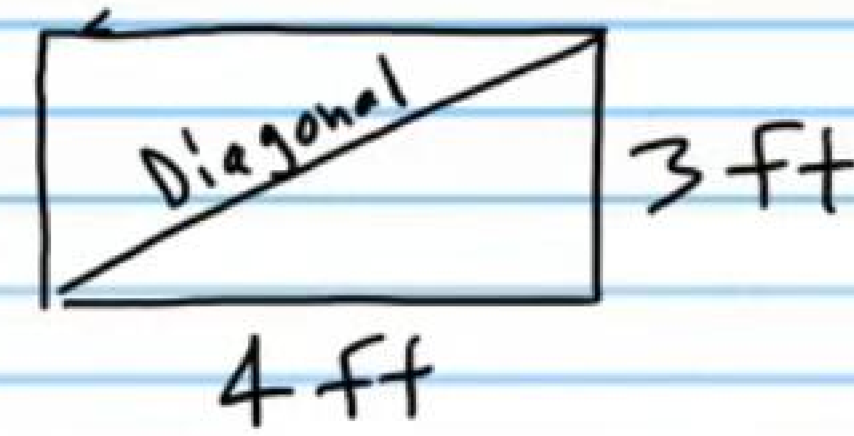
Find the missing side.

(18)



$\boxed{x = 6}$

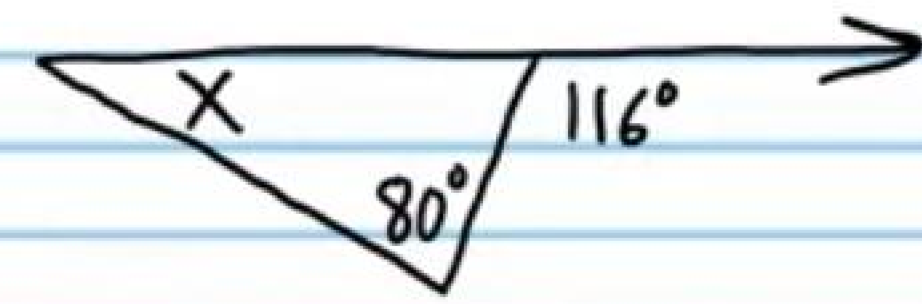
(19) A box has a width of 4 feet and a height of 3 feet. What is the diagonal length of the front of the box?



$\boxed{5\text{ ft}}$

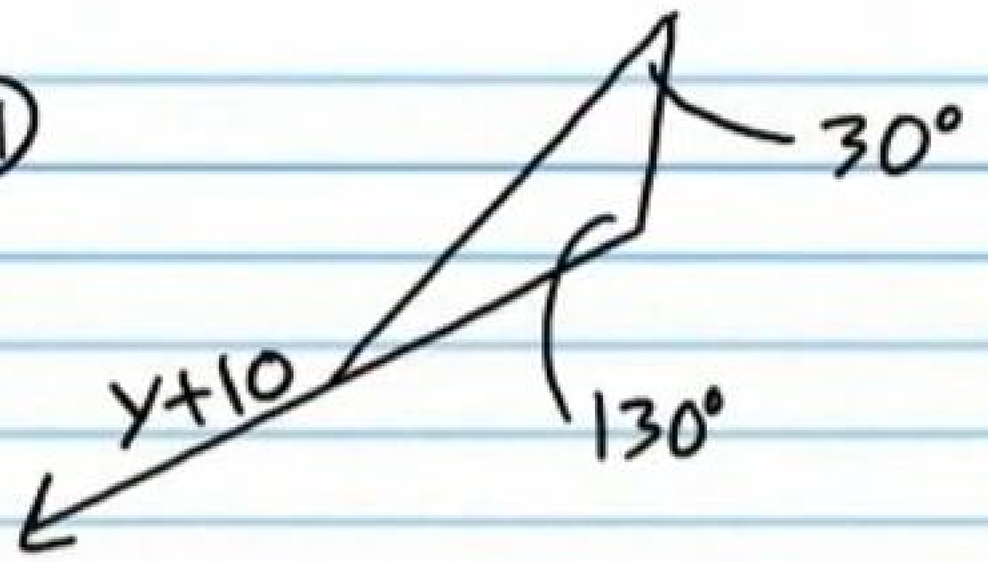
Find x.

(20)



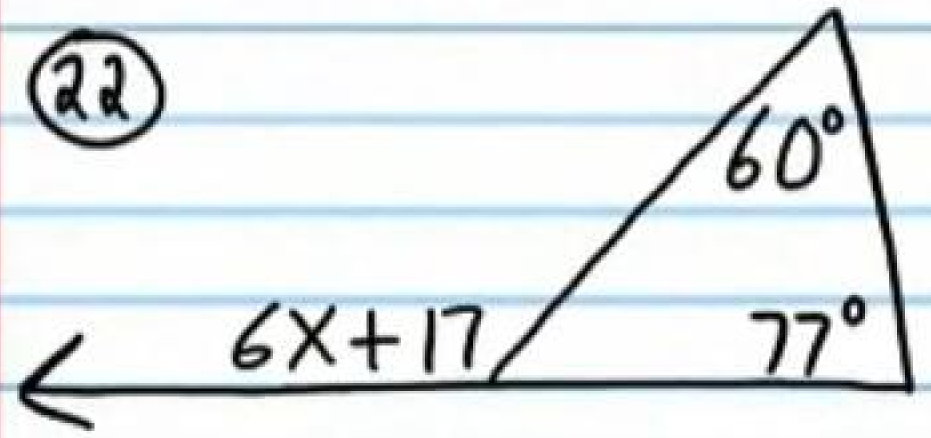
$\boxed{x = 36°}$

(21)



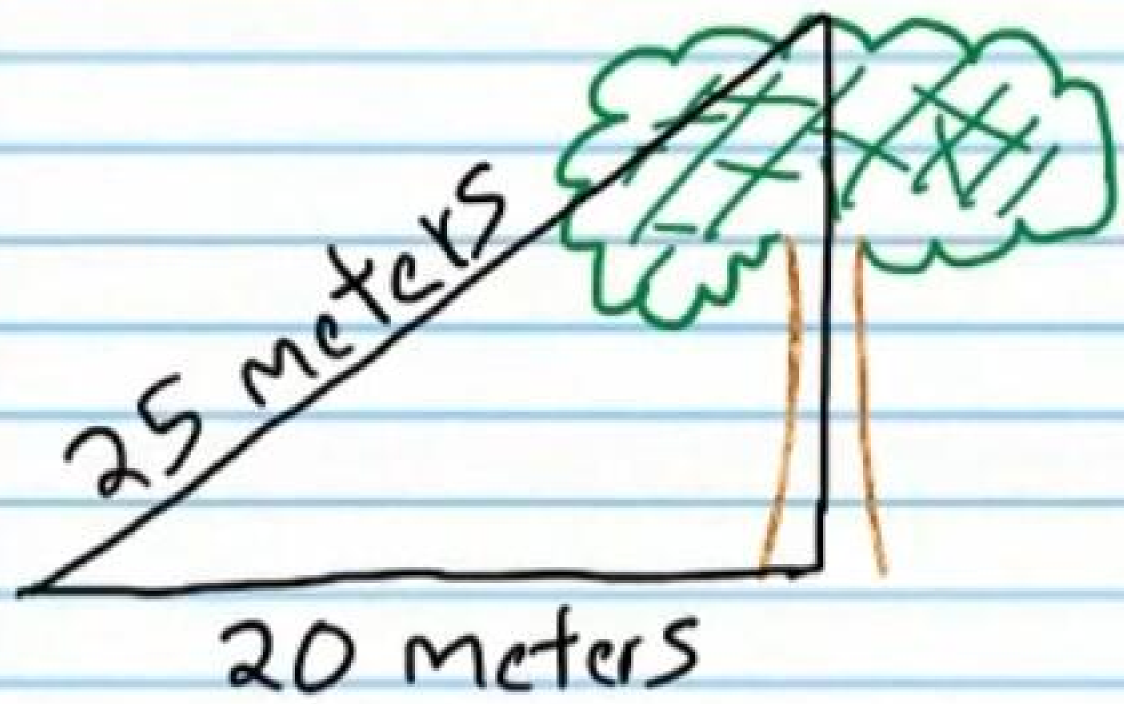
$\boxed{y = 150}$

(22)



$\boxed{X = 20}$

(23) Given the dimensions below, find the height of the tree.



$\boxed{15\text{ M}}$

(24) Write a formal two-column proof to show that the following statement is true: If all three interior angles in the triangle below are congruent, then the measure of each angle is 60°. Hint: You must use the Triangle Sum Theorem. You can say that the given information is the figure.

Answer to Problem # 24

	Statement	Reason
I)	Figure.	Given.
II)	$X + X + X = 180°$	Δ Sum Thm.
III)	$3X = 180°$	Simplification.
IV)	$X = 60°$	Div. Prop. of =.